Normal Distribution

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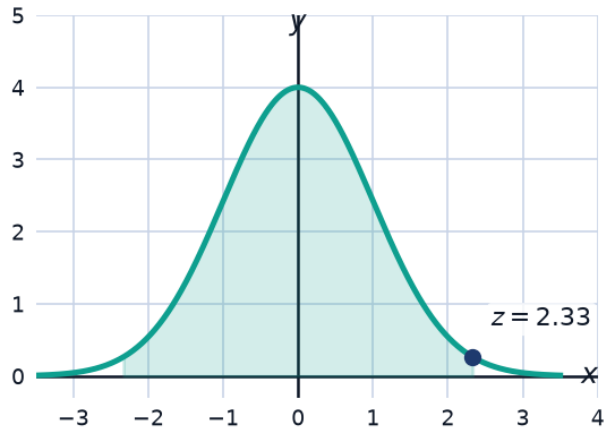
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Properties of the normal distribution

State two key properties of the normal distribution curve.

2 [Foundational] • Target: Properties of the normal distribution

Explain what it means for the normal distribution curve to be "asymptotic" to the horizontal axis.



3 [Foundational] • Target: Properties of the normal distribution

State one similarity and one difference between the normal distribution and the discrete uniform distribution.

4 [Foundational] • Target: Standardizing a normal variable

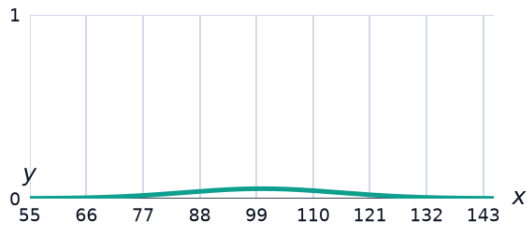
A random variable $X \sim N(50, 64)$. Find the z-score corresponding to $x = 62$, using $z = \frac{x - \mu}{\sigma}$.

5 [Foundational] • Target: Standardizing a normal variable

A random variable $X \sim N(30, 25)$. Find the z-score corresponding to $x = 23$.

6 [Foundational] • Target: Finding probabilities using the standard normal table

A random variable $X \sim N(100, 225)$. Find $P(X < 115)$, given that $\Phi(1) \approx 0.8413$.



7 [Foundational] • Target: Finding probabilities using the standard normal table

Using the same distribution $X \sim N(100, 225)$, find $P(X > 130)$, given that $\Phi(2) \approx 0.9772$.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Finding a value given a probability (inverse normal)**

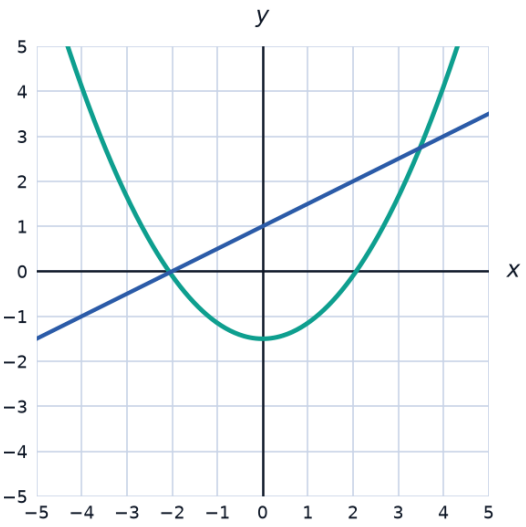
A random variable $X \sim N(60, 100)$. Find the value x such that $P(X < x) = 0.9332$, given that $\Phi(1.5) \approx 0.9332$.

9 **[Intermediate]** • **Target: The normal approximation to the binomial distribution**

A random variable $X \sim B(200, 0.4)$ is to be approximated by a normal distribution. State the mean and variance to use, and check the approximation is valid (both np and $n(1-p)$ should exceed 5).

10 **[Intermediate]** • **Target: The normal approximation to the binomial distribution**

A random variable $X \sim B(150, 0.5)$ is to be approximated by a normal distribution. State the mean and variance to use, and check the approximation is valid.



11 **[Intermediate]** • **Target: Solving for unknown parameters of a normal distribution**

A random variable $X \sim N(\mu, 36)$ satisfies $P(X < 50) = 0.8413$, given $\Phi(1) \approx 0.8413$. Find μ.

12 **[Intermediate]** • **Target: A well-designed word problem: setting a pass mark for an exam**

This question has two parts. Scores on a standardized exam are normally distributed with mean $\mu = 68$ and standard deviation $\sigma = 12$. The exam board wants to set a pass mark so that exactly the top 10% of students pass, and is told that the z-value corresponding to the top 10% (i.e. $P(Z > z) = 0.10$) is $z \approx 1.2816$. (a) Find the pass mark, to the nearest whole number. (b) A student scores 90 on the exam. Find their z-score, and state whether they are in the top 10%.

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13 [Intermediate] • Target: The probability of a value lying between two bounds

Practice 3.2: Ship A leaves port P on a bearing of 040° traveling at 15 km/h. At the same time, ship B leaves P on a bearing of 130° traveling at 20 km/h. Find the distance between the two ships after 2 hours.

14 [Intermediate] • Target: The normal approximation to the binomial distribution

Cars arrive at a toll booth at an average rate of 3 per minute. Find the probability that (a) exactly 5 arrive in one minute, (b) at least 2 arrive in 30 seconds.

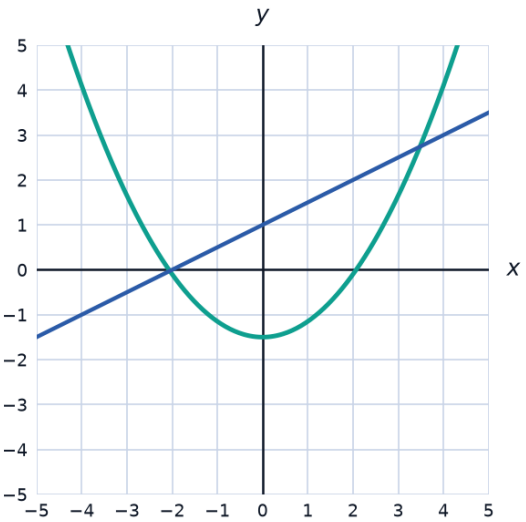
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Solving for unknown parameters of a normal distribution • Cross-Lesson: 15.01 Representation of Data

[Poisson Distribution · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

16 [Challenging] • Target: A well-designed word problem: setting a pass mark for an exam • Cross-Lesson: 16.01 Permutations & Combinations

[Polynomials, Remainder Theorem & Binomial Theorem · Practice 1] If the governing relation for this system is $g(t) = 3t + 3$, evaluate $g(4)$ and find the inverse input t when $g(t) = 18$.



17 [Challenging] • Target: Properties of the normal distribution • Cross-Lesson: 17.01 Discrete Random Variables

[Vectors in Two Dimensions & Relative Motion · Practice 1] If the governing relation for this system is $g(t) = 3t + 3$, evaluate $g(4)$ and find the inverse input t when $g(t) = 18$.

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18 [Challenging] • Target: Standardizing a normal variable • Cross-Lesson: 18.01 Binomial Distribution

[Sequences, Series, Induction & the Binomial Theorem · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

19 [Challenging] • Target: Finding probabilities using the standard normal table • Cross-Lesson: 2.01 Simultaneous Equations & Inequalities

In a factory, 60% of components are manufactured by Machine A and 40% by Machine B. It is known that 3% of components produced by Machine A are defective, whereas 5% of components produced by Machine B are defective. (a) Find the probability that a randomly chosen component is defective. (b) Given that a randomly chosen component is defective, find the probability that it was produced by Machine B. (c) Determine whether the events "component produced by Machine B" and "component is defective" are independent.

20 [Challenging] • Target: Finding a value given a probability (inverse normal) • Cross-Lesson: 3.01 Graphs & Transformations

Given vectors $\mathbf{a} = 2\mathbf{i} - 5\mathbf{j}$ and $\mathbf{b} = -3\mathbf{i} + 4\mathbf{j}$, find the scalar projection of $\mathbf{a}$ onto $\mathbf{b}$ and the angle θ between them to the nearest tenth of a degree.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Properties of the normal distribution

The histogram below shows the number of hours 100 students spent studying for an exam. The number of students who studied for between 0 and 5 hours is represented by a bar with a height of 30. Determine the number of students who studied for between 5 and 10 hours.

a. 20

b. 25

c. 30

d. 35

22 [Foundational] • Target: Standardizing a normal variable

The following two-way table shows the results of a survey on favourite sports among a group of students. Calculate the total number of students surveyed.

a. 50

b. 45

c. 60

d. 55

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23 [Foundational] • Target: Finding probabilities using the standard normal table

Normal distribution $X \sim N(50, 9)$: P(X<50)=

a. 0

b. 0.25

c. 0.5

d. 1

24 [Foundational] • Target: Finding a value given a probability (inverse normal)

Two circles, C_1 with centre O_1 and radius 6 cm, and C_2 with centre O_2 and radius $6\sqrt{3}$ cm, intersect at the points A and B. The distance between their centres is $O_1O_2 = 12$ cm. Find the exact area of the region common to both circles, in cm^2.

a. $30\pi - 36\sqrt{3}$

b. $30\pi + 36\sqrt{3}$

c. $42\pi - 36\sqrt{3}$

d. $15\pi - 18\sqrt{3}$

25 [Foundational] • Target: The probability of a value lying between two bounds

A straight line passing through the origin is tangent to both the curve C_1 with equation $y = e^{kx}$, where k is a positive constant, and the curve C_2 with equation $y = \ln x$. Determine the exact distance between the two points of contact.

a. $(e-1)\sqrt{e^2+1}$

b. $e\sqrt{e^2+1}$

c. $(e-1)\sqrt{e^2-1}$

d. $\sqrt{e^4-e^2+1}$

26 [Foundational] • Target: The normal approximation to the binomial distribution

In a Voronoi diagram, two sites are located at $A(2, 4)$ and $B(6, 2)$. Which of the following is the equation of the boundary edge lying on the perpendicular bisector of the line segment AB?

a. $y = 2x - 5$

b. $y = -\frac{1}{2}x + 5$

c. $y = 2x - 8$

d. $y = -2x + 11$

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27 [Foundational] • Target: Solving for unknown parameters of a normal distribution

In triangle ABC, point D lies on side BC such that AD is an angle bisector of angle A. Let $AB = c$, $AC = b$, $BD = m$, and $CD = n$. Prove that $\frac{m}{n} = \frac{c}{b}$ using a combination of the Sine Rule and the properties of angle bisectors.

a. This statement is a direct consequence of the Sine Rule applied to triangles ABD and ACD.

b. This statement can be proven by applying the Law of Cosines to triangles ABD and ACD and equating expressions for AD^2.

c. This statement is true only if triangle ABC is isosceles with $b = c$.

d. This statement requires the use of vector methods for its proof.

28 [Foundational] • Target: A well-designed word problem: setting a pass mark for an exam

The histogram below shows the distribution of scores for a class of students on a recent quiz. The total number of students who took the quiz is 50. The frequency for the score interval [70, 80) is x. Determine the value of x.

a. 5

b. 10

c. 15

d. 20

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Properties of the normal distribution

Find the standard deviation of the data set: $12, 19, 14, 22, 24$, correct to 2 decimal places.

a. 4.98

b. 4.58

c. 4.18

d. 3.78

30 [Intermediate] • Target: Standardizing a normal variable

A bag contains 8 red and 9 blue balls. Two balls are drawn at random without replacement. Find $P(\text{both red})$.

a. $\frac{64}{289}$

b. $\frac{8}{17}$

c. $\frac{7}{34}$

d. $\frac{56}{289}$

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31 [Intermediate] • Target: Finding probabilities using the standard normal table

Solve $|x - 6| > 6$. What is the smaller boundary value separating the two regions?

a. 11

b. 0

c. 12

d. 1

32 [Intermediate] • Target: Finding a value given a probability (inverse normal)

Solve $|2x + 3| = 3$ for x. What is the larger of the two solutions?

a. 0

b. -3

c. 3

d. $\frac{1}{12}$

33 [Intermediate] • Target: The probability of a value lying between two bounds

Two lines have direction vectors $\mathbf{d_1} = (-2, 3, -5)$ and $\mathbf{d_2} = (4, -6, 10)$. What is the relationship between the two lines (assuming they are coplanar)?

a. neither parallel nor perpendicular

b. perpendicular

c. identical

d. parallel

34 [Intermediate] • Target: The normal approximation to the binomial distribution

Expand $\log\left(\frac{x}{y}\right)$ using properties of logarithms. (variant 6)

a. $\frac{\log x}{\log y}$

b. $\log(x) \cdot \log(y)$

c. $\log x + \log y$

d. $\log x - \log y$

35 [Intermediate] • Target: Solving for unknown parameters of a normal distribution

Expand $\log\left(\frac{x^2}{y^3}\right)$ using properties of logarithms.

a. $\log(x) \cdot \log(y)$

b. $2\log x - 3\log y$

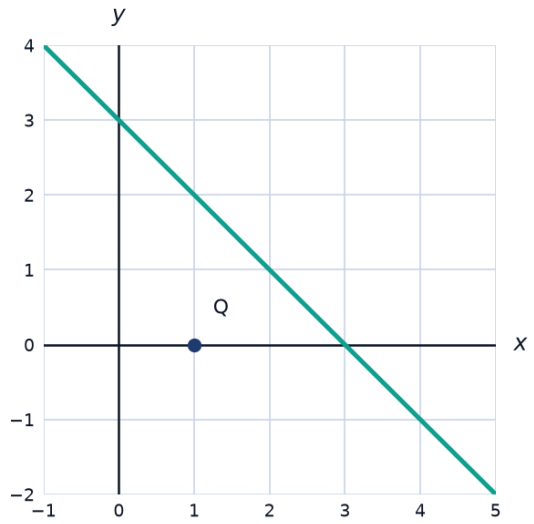
c. $2\log x + 3\log y$

d. $\frac{\log x}{\log y}$

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36 **[Intermediate]** • **Target: A well-designed word problem: setting a pass mark for an exam**

Consider the two functions $f(x) = \log_2(x)$ and $g(x) = 3 - x$. The graphs of $f(x)$ and $g(x)$ intersect at a point P, and $f(x)$ intersects the x-axis at point Q. A line segment is drawn from Q to a point R on the graph of $g(x)$ such that QR is perpendicular to the x-axis. Find the coordinates of the point R.



a. (3, 0)

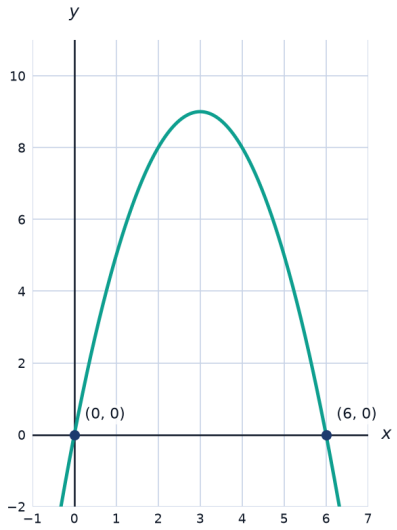
b. (1, 2)

c. (2, 1)

d. (4, -1)

37 **[Intermediate]** • **Target: Properties of the normal distribution**

The diagram shows part of the curve $y = 6x - x^2$, which intersects the x-axis at $(0, 0)$ and $(6, 0)$. Find the area of the region enclosed between the curve and the x-axis.



a. 27

b. 36

c. 54

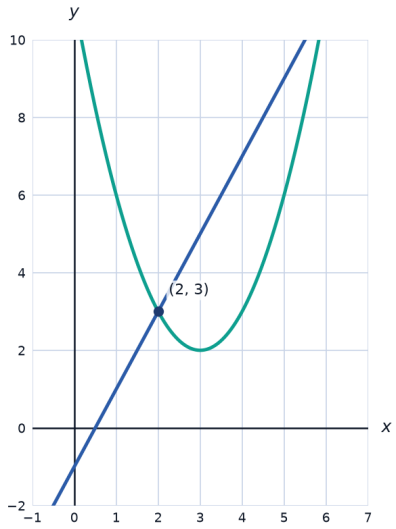
d. 72

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Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 **[Challenging]** • **Target: Standardizing a normal variable** • **Cross-Lesson: 18.01 Binomial Distribution**

Consider the functions $f(x) = x^2 - 6x + 11$ and $g(x) = 2x - 1$. The graphs of these functions intersect at two points. One of the intersection points is $(2, 3)$. Find the x-coordinate of the other intersection point.



a. 3

b. 4

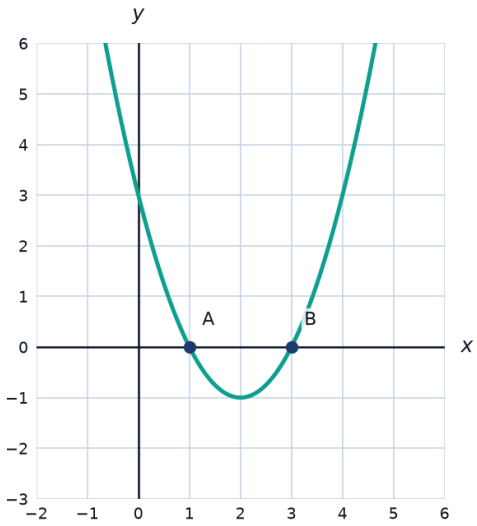
c. 5

d. 6

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39 **[Challenging] • Target: Finding probabilities using the standard normal table • Cross-Lesson: 2.01 Simultaneous Equations & Inequalities**

Consider the quadratic function $f(x) = x^2 - 4x + 3$. The graph of this function is shown below. The points where the graph intersects the x-axis are denoted by A and B. What is the distance between points A and B?



a. 1

b. 2

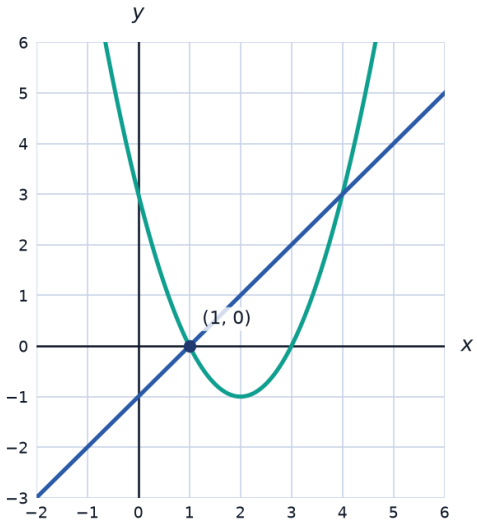
c. 3

d. 4

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40 [Challenging] • Target: Finding a value given a probability (inverse normal) • Cross-Lesson: 3.01 Graphs & Transformations

Consider the function $f(x) = x^2 - 4x + 3$. The graph of $y = f(x)$ is shown below. The function intersects the x-axis at two points. Find the coordinates of the point with the smaller x-value.



a. (1, 0)

b. (3, 0)

c. (0, 3)

d. (0, 1)

41 [Challenging] • Target: The probability of a value lying between two bounds • Cross-Lesson: 4.01 Coordinate Geometry

A curve has equation $y = x\sqrt{16 - x^2}$ for $0 \le x \le 4$. A point $P(x, y)$ moves along the curve in the first quadrant, and the normal to the curve at P intersects the x-axis at the point N. Given that the x-coordinate of P is increasing at a constant rate of $\sqrt{3}$ units s^{-1}, find the rate of change of the area of triangle OPN, where O is the origin, at the instant when $x = 2$.

a. -6 units^2 s^{-1}

b. 6 units^2 s^{-1}

c. -12 units^2 s^{-1}

d. $-2\sqrt{3}$ units^2 s^{-1}

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42 [Challenging] • Target: The normal approximation to the binomial distribution • Cross-Lesson: 5.01 Trigonometry

The curve C has equation $y = x^3 - 3x + 2$. The tangent to C at the point $P(2, 4)$ intersects the curve again at point Q. Find the area of the finite region completely enclosed between the curve C and the tangent line.

a. 108

b. 84

c. 64

d. 12

43 [Challenging] • Target: Solving for unknown parameters of a normal distribution • Cross-Lesson: 6.01 Differentiation & Integration

Consider the two functions $f(x) = ax^2 + bx + c$ and $g(x) = dx + e$. The graph of $f(x)$ is a parabola with its vertex at $(3, -1)$ and passes through the point $(1, 3)$. The graph of $g(x)$ is a straight line passing through the points $(2, 5)$ and $(6, -3)$. If the graphs of $f(x)$ and $g(x)$ intersect at two points, and the x-coordinate of one of these intersection points is p, find the value of $\frac{a+d}{p}$ given that p is an integer.

a. 3

b. 2

c. -2

d. -3

44 [Challenging] • Target: A well-designed word problem: setting a pass mark for an exam • Cross-Lesson: 7.01 Statistics & Probability

Two parallel lines, L_1 and L_2, have n and $n + 2$ distinct points marked on them respectively, where n is an integer with $n \geq 2$. Triangles are formed by choosing 3 of these points as vertices such that they are not collinear. Quadrilaterals are formed by choosing 4 of these points as vertices such that no three are collinear. Given that the ratio of the number of quadrilaterals to the number of triangles that can be formed is $6 : 5$, find the total number of triangles that can be formed.

a. 175

b. 210

c. 220

d. 96

45 [Challenging] • Target: Properties of the normal distribution • Cross-Lesson: 8.01 Functions: Composite & Inverse

The region R in the first quadrant is bounded by the curve $y = \frac{6}{\sqrt{3x+4}}$, the coordinate axes, and the vertical line $x = k$, where $k > 0$. Given that the total area of R is 8, and the vertical line $x = c$ (where $0 < c < k$) divides R into two regions of equal area, find the exact value of c.

a. $\frac{5}{3}$

b. 2

c. $\frac{7}{3}$

d. $\frac{13}{27}$

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